413/613 Fall 2026 - Applied Statistics & Data Science I - Lecture 12

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Recall

In the previous lecture, we considered the problem of estimating an unknown parameter θ from observed data

$$\mathbf{x} = (x_1, \ldots, x_n).$$

Maximum likelihood estimation chooses the parameter value that maximizes the likelihood of the observed data:

$$\boxed{\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} p(\mathbf{x} \mid \theta).}$$

In Bayesian inference, we additionally specify a prior distribution $p(\theta)$ and use Bayes' rule to obtain the posterior distribution

$$\boxed{p(\theta \mid \mathbf{x}) \propto p(\mathbf{x} \mid \theta) p(\theta).}$$

For example, suppose

$$Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2).$$

The unknown parameters are

$$\theta = (\mu, \sigma^2),$$

and they are estimated from the observed sample

$$\mathbf{y} = (y_1, \ldots, y_n).$$

This model specifies the marginal distribution of the response variable Y.

Now suppose that each response Y_i is observed together with a predictor X_i. The data are therefore pairs

$$(x_1, y_1), \ldots, (x_n, y_n).$$

We now model the conditional distribution of the response given the predictor:

$$Y \mid X = x \sim \mathcal{N}\big(\mu(x), \sigma^2\big)\,.$$

Its conditional mean is

$$\mu(x) = \mathbb{E}[Y \mid X = x].$$

The regression problem is therefore to determine how the conditional mean

$$\mathbb{E}[Y \mid X = x]$$

depends on the predictor value x.

Primer: Multivariate Random Variables and Gaussian Distributions

For a random vector

$$\mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_d \end{pmatrix} \in \mathbb{R}^{d\times 1},$$

the mean vector and covariance matrix are

$$\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}] \in \mathbb{R}^{d\times 1},$$

and

$$\Sigma = \mathrm{Cov}(\mathbf{X}) = \mathbb{E}\left[(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^T\right] \in \mathbb{R}^{d\times d}.$$

The entries of Σ are

$$\Sigma_{jk} = \mathrm{Cov}(X_j, X_k),$$

so the diagonal entries are variances and the off-diagonal entries are covariances. A multivariate Gaussian is written

$$\boxed{\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)}$$

with density

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\right].$$

For a multivariate Gaussian, $\boldsymbol{\mu}$ and Σ completely determine the joint distribution.

1 From Estimating a Mean to Estimating a Conditional Mean

Suppose Y is a random variable and we want to predict a future realization of Y by a single number a. Under squared-error loss, the expected prediction error is

$$\mathbb{E}\left[(Y-a)^2\right].$$

The value of a that minimizes this quantity is

$$\boxed{a^* = \mathbb{E}[Y].}$$

Thus, if no additional information is available, the optimal squared-error predictor is the population mean.

Now suppose that before predicting Y, we observe another random variable X. We may then allow the prediction to depend on the observed value of X. Let

$$g(X)$$

denote such a predictor. We seek the function g that minimizes

$$\mathbb{E}\left[(Y-g(X))^2\right].$$

For a fixed value $X = x$, this becomes

$$\mathbb{E}\left[(Y-g(x))^2 \mid X = x\right].$$

The minimizing value is

$$g^*(x) = \mathbb{E}[Y \mid X = x].$$

We therefore define the regression function

$$\mu(x) = \mathbb{E}[Y \mid X = x].$$

The random variable Y can then be decomposed as

$$Y = \mu(X) + \varepsilon,$$

where

$$\varepsilon = Y - \mu(X).$$

Since $\mu(X) = \mathbb{E}[Y \mid X]$,

$$\begin{aligned}\mathbb{E}[\varepsilon \mid X] &= \mathbb{E}[Y - \mu(X) \mid X] \\ &= \mathbb{E}[Y \mid X] - \mu(X) = 0.\end{aligned}$$

Hence,

$$\mathbb{E}[\varepsilon \mid X] = 0.$$

Regression therefore concerns estimating the conditional mean function

$$\mu(x) = \mathbb{E}[Y \mid X = x].$$

2 Simple Linear Regression

The regression function $\mu(x)$ is generally unknown. In simple linear regression, we restrict it to the class of linear functions

$$\mu(x) = \beta_0 + \beta_1 x,$$

where β_0 and β_1 are unknown population parameters.
Since

$$\mu(X) = \mathbb{E}[Y \mid X],$$

the simple linear regression model is

$$Y = \beta_0 + \beta_1 X + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid X] = 0.$$

Equivalently,

$$\mathbb{E}[Y \mid X = x] = \beta_0 + \beta_1 x.$$

Suppose we observe n predictor–response pairs

$$(x_1, y_1), \ldots, (x_n, y_n).$$

For a given choice of parameters β_0 and β_1, the model value at x_i is

$$\beta_0 + \beta_1 x_i.$$

The discrepancy between the observed response y_i and this model value is

$$r_i(\beta_0, \beta_1) = y_i - \beta_0 - \beta_1 x_i.$$

The parameters β_0 and β_1 are unknown. The next question is therefore

How should we estimate β_0 and β_1 from the observed data?

Ordinary least squares answers this by choosing the parameter values that minimize the sum of squared discrepancies.

3 Ordinary Least Squares

Suppose we observe n predictor–response pairs

$$(x_1, y_1), \ldots, (x_n, y_n),$$

and use the linear model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

For given values of β_0 and β_1, the model predicts

$$\hat{y}_i(\beta_0, \beta_1) = \beta_0 + \beta_1 x_i.$$

The corresponding residual is the difference between the observed and predicted response:

$$r_i(\beta_0, \beta_1) = y_i - \beta_0 - \beta_1 x_i.$$

Ordinary least squares chooses β_0 and β_1 to minimize the sum of squared residuals,

$$J(\beta_0, \beta_1) = \sum_{i=1}^{n} r_i(\beta_0, \beta_1)^2 = \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 x_i\right)^2.$$

The least-squares estimates are therefore

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg\min_{\beta_0, \beta_1} J(\beta_0, \beta_1).$$

Since J is a quadratic function of β_0 and β_1, its minimum is obtained by setting both partial derivatives equal to zero. For β_0,

$$\begin{aligned}\frac{\partial J}{\partial \beta_0} &= \sum_{i=1}^{n} 2\left(y_i - \beta_0 - \beta_1 x_i\right)(-1) \\ &= -2\sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 x_i\right).\end{aligned}$$

Hence, at the least-squares solution,

$$\sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right) = 0.$$

If we define the fitted residuals

$$e_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i,$$

then

$$\sum_{i=1}^{n} e_i = 0. \tag{1}$$

For β_1,

$$\begin{aligned}\frac{\partial J}{\partial \beta_1} &= \sum_{i=1}^{n} 2\left(y_i - \beta_0 - \beta_1 x_i\right)(-x_i) \\ &= -2\sum_{i=1}^{n} x_i \left(y_i - \beta_0 - \beta_1 x_i\right).\end{aligned}$$

Therefore,

$$\sum_{i=1}^{n} x_i e_i = 0. \tag{2}$$

These two equations Eq. (1) and (2) are the **normal equations** for simple linear regression. From

$$\sum_{i=1}^{n} e_i = 0,$$

we obtain

$$\sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i \right) = 0.$$

Dividing by n gives

$$\bar{y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{x} = 0,$$

so

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Substituting this into the second normal equation gives

$$\sum_{i=1}^{n} x_i \left[(y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x}) \right] = 0.$$

Using

$$\sum_{i=1}^{n} (y_i - \bar{y}) = 0, \quad \sum_{i=1}^{n} (x_i - \bar{x}) = 0,$$

this can be written as

$$\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = \hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

Therefore,

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}.$$

Finally,

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Thus the fitted regression line is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

3.1 Why Squared Error? Return to Maximum Likelihood

In the earlier Gaussian estimation problem, we assumed

$$Y_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2), \qquad i = 1, \ldots, n,$$

and estimated the unknown parameters μ and σ^2 from the observed responses

$$\mathbf{y} = (y_1, \ldots, y_n).$$

In regression, each response Y_i is observed together with a predictor value x_i. We now specify the conditional distribution of Y_i given $X_i = x_i$ as

$$Y_i \mid X_i = x_i \sim \mathcal{N}\left(\beta_0 + \beta_1 x_i, \sigma^2 \right).$$

Thus, for observation i,

$$\mathbb{E}[Y_i \mid X_i = x_i] = \beta_0 + \beta_1 x_i,$$

and

$$\mathrm{Var}(Y_i \mid X_i = x_i) = \sigma^2.$$

Equivalently,

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

where

$$\varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2).$$

For a given predictor value x_i, the conditional density of the observed response y_i is therefore

$$p(y_i \mid x_i, \beta_0, \beta_1, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right].$$

Assuming the responses are conditionally independent given the predictor values $x_1, \ldots, x_n$, the likelihood of the observed response vector

$$\mathbf{y} = (y_1, \ldots, y_n)$$

is

$$\begin{aligned} L(\beta_0, \beta_1, \sigma^2) &= p(\mathbf{y} \mid \mathbf{x}, \beta_0, \beta_1, \sigma^2) \\ &= \prod_{i=1}^{n} p(y_i \mid x_i, \beta_0, \beta_1, \sigma^2) \\ &= (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2\right]. \end{aligned}$$

Taking logarithms gives

$$\ell(\beta_0, \beta_1, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2.$$

If σ^2 is fixed, the first two terms do not depend on β_0 or β_1. Therefore,

$$\arg\max_{\beta_0, \beta_1} \ell(\beta_0, \beta_1, \sigma^2) = \arg\min_{\beta_0, \beta_1} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2.$$

The squared residuals therefore arise directly from the Gaussian error model.

4 Linear Regression in Matrix Form

For simple linear regression, we observe n pairs

$$(x_1, y_1), \ldots, (x_n, y_n),$$

and model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \ldots, n.$$

Collect the responses into the vector

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^{n \times 1},$$

and define the design matrix

$$X = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \in \mathbb{R}^{n\times 2}.$$

The coefficient vector is

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} \in \mathbb{R}^{2\times 1},$$

and the error vector is

$$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix} \in \mathbb{R}^{n\times 1}.$$

Since

$$X\beta \in \mathbb{R}^{n\times 2}\mathbb{R}^{2\times 1} = \mathbb{R}^{n\times 1},$$

the n scalar regression equations can be written compactly as

$$\boxed{\mathbf{y} = X\beta + \varepsilon.}$$

Indeed, the ith entry of $X\beta$ is

$$\begin{pmatrix} 1 & x_i \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \beta_0 + \beta_1 x_i.$$

4.1 Residual vector

For a candidate coefficient vector β, define

$$\mathbf{r} = \mathbf{y} - X\beta \in \mathbb{R}^{n\times 1}.$$

Its ith entry is

$$r_i = y_i - \beta_0 - \beta_1 x_i.$$

The residual sum of squares is therefore

$$J(\beta) = \sum_{i=1}^{n} r_i^2.$$

In matrix notation,

$$\boxed{J(\beta) = \mathbf{r}^T\mathbf{r} = (\mathbf{y} - X\beta)^T(\mathbf{y} - X\beta).}$$

The dimensions are

$$\mathbf{r}^T\mathbf{r}: \qquad (1\times n)(n\times 1) = 1\times 1,$$

so $J(\beta)$ is a scalar, as required.

4.2 Least-squares solution

Expand the objective:

$$\begin{aligned} J(\beta) &= (\mathbf{y} - X\beta)^T(\mathbf{y} - X\beta) \\ &= \mathbf{y}^T\mathbf{y} - 2\beta^T X^T\mathbf{y} + \beta^T X^T X\beta. \end{aligned}$$

Differentiating with respect to

$$\beta \in \mathbb{R}^{2\times 1}$$

gives

$$\nabla_\beta J = -2X^T\mathbf{y} + 2X^T X\beta.$$

The gradient has shape

$$\nabla_\beta J \in \mathbb{R}^{2\times 1},$$

since

$$X^T\mathbf{y}: \qquad (2\times n)(n\times 1) = 2\times 1,$$

and

$$X^T X\beta: \qquad (2\times n)(n\times 2)(2\times 1) = 2\times 1.$$

Setting the gradient equal to zero gives the normal equations

$$\boxed{X^T X\hat{\beta} = X^T\mathbf{y}.}$$

Here,

$$X^T X \in \mathbb{R}^{2\times 2}, \qquad X^T\mathbf{y} \in \mathbb{R}^{2\times 1}.$$

If the two columns of X are linearly independent, then $X^T X$ is invertible and

$$\boxed{\hat{\beta} = (X^T X)^{-1}X^T\mathbf{y}.}$$

The dimensions are

$$(2\times 2)(2\times n)(n\times 1) = 2\times 1,$$

consistent with

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix}.$$

5 Multiple Linear Regression

Suppose that for each observation $i = 1, \ldots, n$, we measure a response Y_i together with p predictors,

$$X_{i1}, X_{i2}, \ldots, X_{ip}.$$

For example, if Y_i is a measure of diabetes progression, the predictors may include BMI, blood pressure, age, and other measured variables. We now model the conditional mean of the response given all predictors:

$$\mathbb{E}[Y_i \mid X_{i1} = x_{i1}, \ldots, X_{ip} = x_{ip}] = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}.$$

Equivalently,

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip} + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid X_{i1}, \ldots, X_{ip}] = 0.$$

The coefficient β_j describes how the conditional mean of Y changes with X_j when the other predictors are held fixed.

Collect the n observations into

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^{n \times 1}, \qquad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} \in \mathbb{R}^{(p+1) \times 1}.,$$

and define the design matrix

$$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix} \in \mathbb{R}^{n \times (p+1)}.$$

Then the model is

$$\mathbf{y} = X\beta + \varepsilon.$$

Ordinary least squares chooses $\hat{\beta}$ to minimize

$$\|\mathbf{y} - X\beta\|_2^2,$$

giving, when $X^T X$ is invertible,

$$\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y}.$$

Thus, moving from simple to multiple linear regression does not change the least-squares problem; it only adds additional predictor columns to the design matrix X.

6 Is the Least-Squares Estimator Unbiased?

The least-squares estimator is

$$\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y}.$$

The observed response vector is modeled as

$$\mathbf{y} = X\beta + \varepsilon,$$

where X is the observed design matrix, β is the unknown true coefficient vector, and ε is the random error vector. Substituting the model for $\mathbf{y}$ gives

$$\begin{aligned} \hat{\beta} &= (X^T X)^{-1} X^T (X\beta + \varepsilon) \\ &= (X^T X)^{-1} X^T X\beta + (X^T X)^{-1} X^T \varepsilon \\ &= \beta + (X^T X)^{-1} X^T \varepsilon. \end{aligned}$$

This expression shows explicitly how the estimated coefficients differ from the true coefficients:

$$\hat{\beta} - \beta = (X^T X)^{-1} X^T \varepsilon.$$

We now ask what happens if we repeatedly observe new response vectors generated from the same predictor values X. Conditional on X, the matrix X and the true parameter vector β are fixed, while the error vector ε is random. Therefore,

$$\begin{aligned} \mathbb{E}[\hat{\beta} \mid X] &= \mathbb{E}\left[\beta + (X^T X)^{-1} X^T \varepsilon \mid X\right] \\ &= \beta + (X^T X)^{-1} X^T \mathbb{E}[\varepsilon \mid X]. \end{aligned}$$

Under the regression assumption

$$\mathbb{E}[\varepsilon \mid X] = 0,$$

we obtain

$$\mathbb{E}[\hat{\beta} \mid X] = \beta.$$

Thus, for fixed predictor values X, the least-squares estimator is unbiased:

$$\mathbb{E}[\hat{\beta}_j \mid X] = \beta_j, \qquad j = 0, \ldots, p.$$

In other words, if we repeatedly generated new response data from the same linear model at the same predictor values, the average estimated coefficient would equal the corresponding true coefficient.

6.1 Variance of the Least-Squares Estimator

From the previous section,

$$\hat{\beta} = \beta + (X^T X)^{-1} X^T \varepsilon.$$

Conditional on the observed design matrix X, the only random quantity is the noise vector ε. Therefore,

$$\operatorname{Var}(\hat{\beta} \mid X) = \operatorname{Var}\left((X^T X)^{-1} X^T \varepsilon \mid X\right).$$

Using the identity

$$\operatorname{Var}(AZ) = A \operatorname{Var}(Z) A^T,$$

with

$$A = (X^T X)^{-1} X^T,$$

gives

$$\operatorname{Var}(\hat{\beta} \mid X) = (X^T X)^{-1} X^T \operatorname{Var}(\varepsilon \mid X) X (X^T X)^{-1}.$$

Under the homoscedastic and uncorrelated-error assumption,

$$\operatorname{Var}(\varepsilon \mid X) = \sigma^2 I_n,$$

meaning that each error has variance σ^2 and

$$\operatorname{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0, \qquad i \neq j.$$

Therefore,

$$\begin{aligned} \operatorname{Var}(\hat{\beta} \mid X) &= (X^T X)^{-1} X^T (\sigma^2 I_n) X (X^T X)^{-1} \\ &= \sigma^2 (X^T X)^{-1} X^T X (X^T X)^{-1} \\ &= \boxed{\sigma^2 (X^T X)^{-1}}. \end{aligned}$$

Hence, for coefficient $\hat{\beta}_j$,

$$\operatorname{Var}(\hat{\beta}_j \mid X) = \sigma^2 \left[(X^T X)^{-1}\right]_{jj}.$$

Thus, the sampling variability of each estimated coefficient is determined by the noise variance σ^2 and by the observed predictor values through $X^T X$.